Mitigating Token Homogenization in Token Merging via Source Selection Switching and Merge Embedding

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Abstract

*Vision Transformers (ViTs) achieve strong performance through global self-attention, but their quadratic token complexity limits efficient deployment. Token Merging (ToMe) reduces this cost by averaging similar tokens; however, we show that aggressive merging introduces **Token Homogenization**, in which attention-driving token components are attenuated in deep layers. Our analysis links this issue to disrupted late-layer attention sharpening: image-to-CLS attention gets reduced, suppressing high-response token peaks under heavy reduction.*

*To address this challenge, we propose two complementary techniques: (1) **Source Selection Switching (SSS)**, which uses similarity-based matching in early layers and attention-based protection in deep layers, and (2) **Merge Embedding (ME)**, a lightweight correction added to source tokens before merging to restore attention-driving components. Together, these design choices are directly motivated by our mechanistic findings and preserve deep-layer selectivity under aggressive token reduction. Experiments on ImageNet-1K with DeiT-S show consistent accuracy gains; for example, SSS and SSS+ME improve Top-1 accuracy by +2.4 and +3.7 points over ToMe, respectively, while being 2.1× faster than the full model. Code is available at* https://github.com/ychikawa/SSS_ME

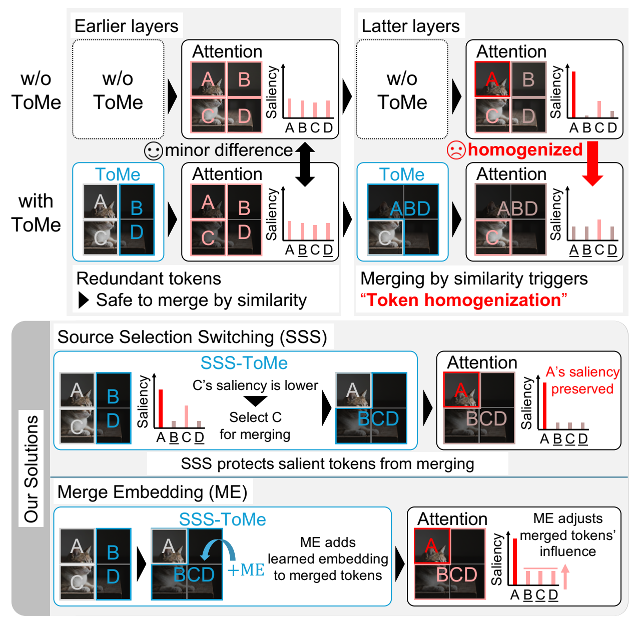


Figure 1. Overview of token homogenization under token merging and our mitigation. Merging attenuates attention-driving components and weakens deep-layer selectivity; Source Selection Switching (SSS) preserves salient anchors, while Merge Embedding (ME) restores missing components to maintain the CLS token's focus on these anchors.

1. Introduction

The Transformer architecture [22] has reshaped computer vision by enabling global reasoning through self-attention. As the primary adaptation of this paradigm to the visual domain, Vision Transformers (ViTs) [5] represent an image as a sequence of patch tokens and apply attention across the entire sequence, often outperforming convolutional models by capturing long-range dependencies. This strength comes at a cost: self-attention compares every token with every other token, so computation and memory grow quadratically with the number of tokens. As image resolution increases, this quickly becomes a bottleneck, limiting real-time or edge deployment scenarios.

To reduce this cost, token reduction methods compress the sequence while preserving accuracy. Prior work explores token pruning and dynamic sparsification [6, 7, 10, 14, 17], token reorganization or pooling [13, 15, 19], and adaptive token schemes [25, 26]. Among them, Token Merging (ToMe) [1] is especially attractive because it combines similar tokens without discarding them and can be applied to pre-trained models without extensive retraining. Intuitively, merging replaces groups of redundant tokens with a single averaged representative, reducing sequence length and accelerating attention by simplifying the token distribution.

While token merging reduces computational cost, it fundamentally alters attention dynamics, thereby compromising accuracy. In standard ViTs, deep layers exhibit *attention sharpening*: the model progressively concentrates its representation on a small set of salient tokens that serve as semantic anchors for classification. This process is exacerbated as the merging becomes more aggressive. By averaging informative tokens with redundant ones, merging attenuates the distinctive components of these anchors, suppressing the QK response peaks that the CLS token relies on. Consequently, image-to-CLS attention weakens in late layers, causing the model to lose focus on the discriminative visual cues. We term this failure mode *Token Homogenization*.

To address this, we propose two complementary techniques, as shown in Figure 1. Source Selection Switching (SSS) adapts the merging criterion by depth: similarity-based matching is applied in early layers, where attention is diffuse and redundancy is high, while attention-based selection is applied in deeper layers, where it protects stable, high-importance semantic anchors. Merge Embedding (ME) complements SSS by adding a lightweight learned correction to merged tokens, restoring the attenuated attention-driving components and stabilizing merged-token influence. Together, SSS and ME preserve deep-layer selectivity under aggressive token reduction.

Our primary contributions are as follows:

- We identify **Token Homogenization** as a key side effect of token merging, where attention-driving token components are attenuated, degrading deep-layer selectivity.
- We analyze attention sharpening and show that increased logit variance (dominated by a bias-dependent term) drives entropy reduction with layer depth, explaining why repeated averaging disrupts late-layer attention behavior.
- We introduce **Source Selection Switching (SSS)** and **Merge Embedding (ME)** to preserve salient tokens and restore attention-driving components during merging.
- We demonstrate strong empirical gains on ImageNet-1K. On DeiT-S, SSS and SSS+ME improve Top-1 accuracy by +2.4 and +3.7 points over ToMe, respectively, while achieving a 2.1× speedup compared to the full model.

2. Related Work

Efficient Vision Transformers reduce computation by shortening token sequences. Prior work spans dynamic token sparsification [17], adaptive token sampling [6], learned pruning [10], and early-exit or slow–fast variants [25], as well as pooling or reorganization that preserves spatial structure while shrinking sequence length [13–15, 19]. These approaches often require training from scratch or substantial retraining to recover accuracy.

Token merging provides a complementary efficiency path by combining similar tokens instead of dropping them, and it can deliver strong performance with relatively limited retraining or calibration. We first highlight **Bipartite Soft Matching (BSM)** methods, which merge tokens based on feature similarity. ToMe [1] established BSM, and later work largely keeps this matching backbone while exploring where to intervene and how to recover information: some approaches combine pruning with merging [2, 9], while others learn soft merging policies [27] or modify the matching criterion or embedding space [11, 12, 21]. Despite their differences, these approaches still determine merge pairs using BSM-like methods.

On the other hand, **Token Pruning and Squeezing (TPS)** methods select salient tokens via attention or saliency scores. It effectively ranks tokens by importance to determine which tokens are preserved or merged, as exemplified by token pruning and squeezing [23]. DiffRate [3] also falls into this category and proposes to learn the token reduction schedule.

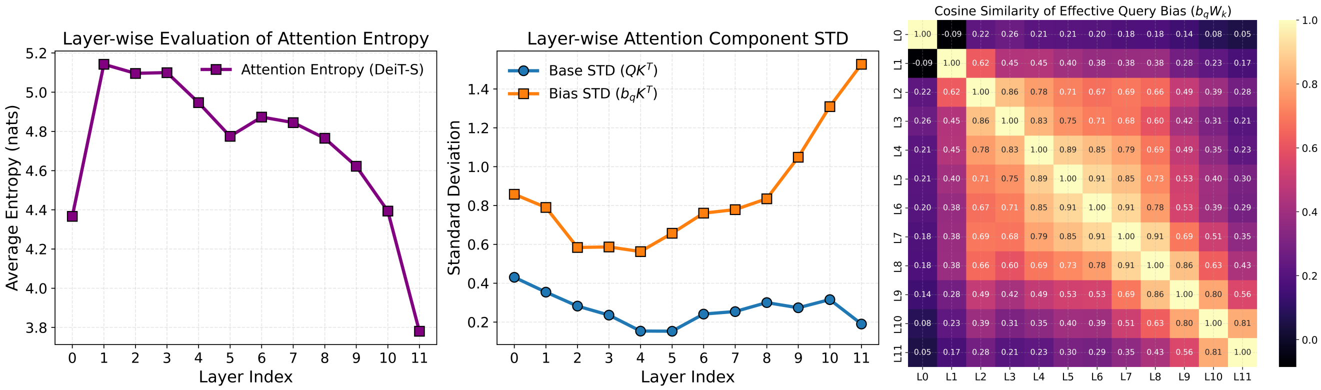
Existing work mainly improves the selection rule or reconstruction after merging. Our work instead analyzes the attention-side effects of merging and introduces depth-aware source selection and merge-time correction to preserve deep-layer selectivity.

3. Analysis of Attention Dynamics and Token Homogenization

In this section, we analyze the internal attention dynamics of Vision Transformers (ViTs) [5] to uncover the mechanistic roots of performance degradation under aggressive token reduction. Using ImageNet-1K validation images (224×224), we examine DeiT-S with 197 tokens (14×14 patches plus a CLS token) [4, 20]. Our analysis reveals a critical failure mode: as token merging progresses, the distinctiveness of image tokens is diluted, causing the CLS token to lose its focus on salient regions and instead collapse toward a high self-attention score. To formalize this, we first quantify the "attention sharpening" observed in late layers and identify the specific logit components that drive this behavior. We then demonstrate how token homogenization disrupts this sharpening process by attenuating attention-driving features. This mechanistic insight directly motivates our two-fold solution in Sec. 4: a depth-aware Source Selection Switching (SSS) and a Merge Embedding (ME) to restore token-wise discriminability.

3.1. Quantifying Attention Sharpening via Entropy

We begin by measuring how attention concentrates with depth using **Average Attention Entropy** per layer l. Let the mean attention probability for key j across all heads H



(a) Layer-wise Entropy (b) Variance Decomposition (c) Cosine Similarity of Bias

Figure 2. Analysis of internal attention dynamics in DeiT-S [20]. (a) Layer-wise transition of average attention entropy, illustrating the sharpening effect. (b) Decomposition of attention score variance into data-dependent and bias-dependent components. (c) Layer-to-layer cosine similarity of the effective query bias, indicating structural consistency in deeper stages.

and query tokens N be

$$\bar{p}_j^{(l)} = \frac{1}{H \cdot N} \sum_{h=1}^{H} \sum_{i=1}^{N} p_{ij}^{(h,l)}, \tag{1}$$

where $p_{ij}^{(h,l)}$ is the attention weight from query i to key j in head h at layer l. We then compute layer-wise entropy as $H^{(l)} = -\sum_{j=1}^{N} \bar{p}_j^{(l)} \log \bar{p}_j^{(l)}$. Lower entropy corresponds to concentrated, consistent focus; higher entropy indicates diffuse attention.

Fig. 2a exhibits a downward trend in entropy with depth in DeiT-S [20], from 5.1 nats at Layer 1 to 3.8 nats at Layer 11. This indicates a transition from broad receptive fields to highly selective, sparse attention, which we call **Attention Sharpening**.

High entropy in *early layers* suggests the absence of stable semantic anchors and a predominance of redundant or background-like tokens. In this regime, similarity-based pairing is comparatively safe because it tends to merge interchangeable background tokens rather than collapsing anchors. This makes similarity-based selection a natural choice for early-layer merging, whereas it is *not* the case for the remaining layers.

The remainder of the section explains the origin of Attention Sharpening and its implications for token merging in the deeper layers.

3.2. Mechanism of Sharpening: Decomposition and Causal Analysis

The reduction in entropy in Fig. 2a implies a structural change in pre-softmax attention logits. We make this connection precise by analyzing how softmax entropy depends on logit variance, then trace the origin of the variance.

Monotonic Relationship between Logit Variance and Entropy Let $\mathbf{z} = (z_1, \ldots, z_N)$ denote the pre-softmax attention logits and

$$p_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \tag{2}$$

the corresponding attention probabilities. We decompose

$$z_i = \mu + \delta_i, \quad \text{with } \mathbb{E}[\delta_i] = 0, \tag{3}$$

and introduce a scaling parameter $\alpha > 0$ to control the logit variance:

$$z_i(\alpha) = \mu + \alpha\delta_i, \tag{4}$$

so that $\mathrm{Var}(\mathbf{z}(\alpha)) = \alpha^2 \mathrm{Var}(\mathbf{z})$.

Let $p_i(\alpha) = \mathrm{softmax}(z_i(\alpha))$ and

$$H(\alpha) = -\sum_i p_i(\alpha) \log p_i(\alpha) \tag{5}$$

be the corresponding entropy. Differentiating with respect to α yields

$$\frac{dH(\alpha)}{d\alpha} = -\alpha \, \mathrm{Var}_{p(\alpha)}(\delta), \tag{6}$$

which implies

$$\frac{dH(\alpha)}{d\alpha} \leq 0. \tag{7}$$

Hence, attention entropy is a monotonically decreasing function of logit variance, and increasing variance necessarily sharpens the distribution.

Dominance of the Bias Term To clarify the cause of attention sharpening, we decompose the pre-softmax attention scores as follows:

$$(x_Q W_Q + b_Q)(x_K W_K + b_K)^\top = z. \tag{8}$$

Expanding this gives four terms: the data interaction $(x_Q W_Q)(x_K W_K)^\top$, two bias-linear terms $b_Q(x_K W_K)^\top$ and $(x_Q W_Q)b_K^\top$, and the constant $b_Q b_K^\top$. Because softmax is invariant to additive shifts, the last two terms do not affect attention weights, leaving only the first two as effective contributors. We therefore decompose the pre-softmax attention logit z into a data-dependent term (*attn_base*) and a bias-dependent term (*attn_bias*) involving the query bias projection:

$$z \approx \underbrace{(x_Q W_Q)(x_K W_K)^\top}_{\text{attn\_base}} + \underbrace{b_Q W_K^\top x_K^\top}_{\text{attn\_bias}}. \tag{9}$$

We track the standard deviation (STD) of both components across layers. As shown in Fig. 2b, the data-dependent term remains relatively stable, whereas the bias-dependent term grows substantially with depth. The *attn_bias* STD rises from about 0.6 at Layer 2 to over 1.5 at Layer 11, becoming the dominant contributor to logit variance.

Combined with the monotonic variance–entropy relationship, this shows that the sharpening in Fig. 2a is primarily driven by *attn_bias*. As $\mathrm{Var}(\text{attn\_bias})$ increasingly outweighs the data-dependent component, the bias term becomes the principal factor reducing entropy and producing sparse attention in deep layers.

Inter-layer Similarity and Token Consistency If the bias term dominates, we also expect its structure to stabilize with depth. We examine this by measuring inter-layer similarity: the effective query bias ($b_Q W_K^\top$) exhibits high cosine similarity among deeper layers in Fig. 2c, indicating convergence to a stable pattern of salient tokens.

To quantify this consistency, we evaluate **Attention Weight Occupancy** across layers. Let $\mathcal{T}_i$ be the set of top-K tokens with the highest attention scores in layer i. We set $K = 30$ (out of 197 tokens) and compute the total attention weight that these tokens receive in layer j as

$$\mathcal{O}_{i,j} = \sum_{k \in \mathcal{T}_i} \bar{p}_k^{(j)}, \tag{10}$$

where $\bar{p}_k^{(j)}$ is the mean attention probability for token k in layer j. As shown in Fig. 3a, salient tokens identified in early-to-mid layers persist into deeper stages. In DeiT-S [20], the top 30 tokens from Layer 6 already account for about 49% of the total attention weight in Layer 11. This occupancy reaches 76% when using Layer 11's own top tokens, confirming convergence to stable "semantic anchors" and providing a concrete basis for late-layer token protection.

Implication for Late-Layer Token Preservation Taken together, dominance of the bias-dependent term and inter-layer token consistency indicate that late-layer CLS attention relies on specific token-wise vector components that define a small set of semantic anchors. When token merging averages token representations, these attention-driving components are attenuated, weakening image-to-CLS interaction and shifting attention toward the CLS token itself.

This suggests that restoring such components at merge time is necessary to maintain late-layer selectivity. A lightweight learned embedding added to merged tokens can compensate for this loss by reintroducing directions that attract the CLS query, motivating our Merge Embedding (ME) as a principled correction for token homogenization.

Implication for Late-Layer Source Selection The same inter-layer consistency also implies that a small set of salient tokens remains reliably identifiable in deep layers. In this regime, similarity-based merging becomes risky because it can collapse these anchors. A more reliable strategy is to select and protect source tokens using attention-based importance scores (e.g., CLS attention), which motivates the late-layer switch in SSS.

3.3. The Side Effect: Token Homogenization

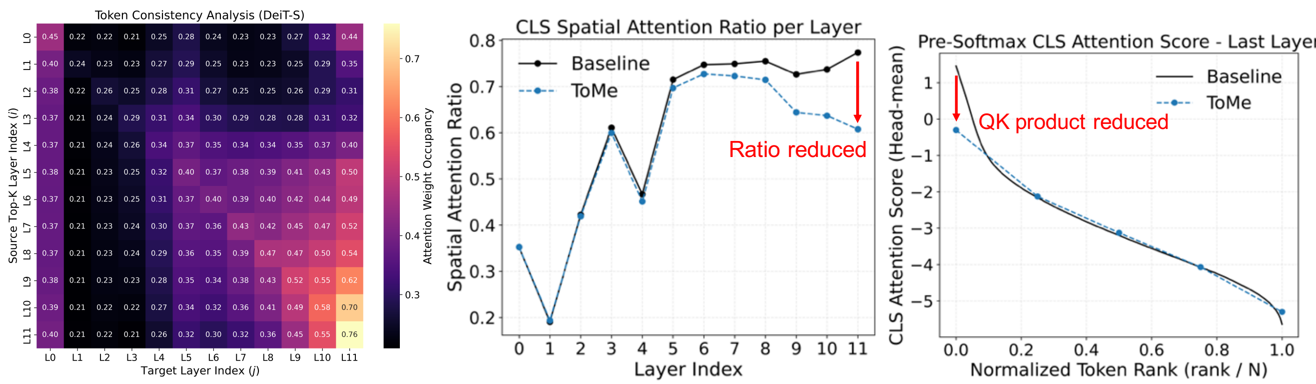
So far we have described the natural sharpening process in standard ViTs [5]. We now show that token merging disrupts this process through **Token Homogenization**: repeated averaging attenuates token-specific components that are required for strong attention from the CLS query, especially in deeper layers where attention is already highly selective.

Reduced CLS Attention to Image Tokens To expose this effect at the probability level, we measure the layer-wise mean of

$$\frac{1}{H}\sum_{h=1}^{H} \sum_{j \in \text{image tokens}} p_{\text{CLS},j}^{(h,l)}, \tag{11}$$

that is, the softmax-normalized CLS attention mass assigned to non-CLS (image) tokens. As shown in Fig. 3b, this quantity decreases in late layers under ToMe, while the full model maintains higher values. This indicates that, with token merging, the CLS token progressively allocates less attention to image tokens.

Suppression of High-Response QK Components We next examine the pre-softmax mechanism using sorted QK inner products in Fig. 3c. In the full model, a subset of tokens exhibits clear high-value peaks. This is consistent with the previous subsection, where tokens containing bias-responsive components receive large late-layer attention. After applying ToMe, these peaks are largely suppressed, indicating that merging weakens the token components that make specific tokens highly responsive to the CLS query.



(a) Token Consistency (b) Image-to-CLS Attention Ratio (c) Pre-softmax QK Inner Product

Figure 3. Quantitative analysis of token consistency and its side effects under token merging in DeiT-S [20]. (a) Attention Weight Occupancy of top-K tokens across layers, indicating that salient tokens persist as semantic anchors. (b) Layer-wise mean attention mass from the image tokens to CLS token after softmax normalization, comparing the full model and ToMe ($r = 20$) [1]. (c) Sorted pre-softmax QK inner products in a deep layer, showing that prominent peaks in the full model are suppressed after applying ToMe ($r = 20$).

Taken together, Figs. 3b and 3c suggest that performance degradation is better explained by **Token Homogenization**: merging smooths token representations, suppresses peak QK responses, and consequently reduces image-to-CLS attention in deep layers. This finding directly motivates the corrective mechanisms in Sec. 4, which are designed to preserve or restore these attention-driving components during merging.

4. Methodology

To address the token homogenization mechanism identified in Sec. 3, we propose two complementary techniques: **Source Selection Switching (SSS)** and **Merge Embedding (ME)**. SSS determines *which* tokens should be merged at each depth, while ME controls *how* merged tokens affect subsequent attention. Together, they preserve deep-layer selectivity and stabilize the contribution of merged tokens with minimal overhead, directly targeting the failure mode revealed by our analysis.

4.1. Source Selection Switching (SSS)

Our analysis in Sec. 3 shows that deeper layers concentrate attention on a small set of salient tokens. As depth increases, these peaks become sharper and more stable. In contrast, early layers exhibit diffuse attention and substantial redundancy. This depth-dependent shift is crucial: a criterion that is effective in early layers can become harmful in later layers, and vice versa.

Existing methods can be grouped into two families, as discussed in Sec. 2. **Bipartite Soft Matching (BSM)** methods merge tokens according to feature similarity (e.g., ToMe and its extensions) [1, 11, 12]. **Token Pruning and Squeezing (TPS)** methods rank tokens by attention or saliency, preserve high-importance tokens, and reduce the rest (e.g., token pruning and squeezing; Diffrate) [3, 23]. These two families are effective in different regimes: BSM performs well when redundancy is high, whereas TPS is better suited to concentrated late-layer attention. The core challenge is that a single criterion across all layers either over-merges salient tokens in deep layers or under-merges redundant tokens in early layers.

To address this mismatch, we introduce **Source Selection Switching (SSS)**, which adapts the merge-selection criterion by depth. Let $l \in \{0, \dots, L-1\}$ be the layer index and d the switch depth. We define the source-selection rule as

$$\text{criterion}(l) = \begin{cases} \text{BSM} & \text{if } l < d, \\ \text{TPS} & \text{otherwise}, \end{cases} \tag{12}$$

where BSM denotes similarity-based bipartite soft matching, and TPS denotes attention/saliency-based token ranking that preserves high-importance tokens (e.g., by CLS attention) while reducing the rest. In practice, this yields the following behavior, as illustrated in Fig. 4:

- **Early Layers**: Use BSM-style matching, which is effective when attention is diffuse and redundancy is high, enabling aggressive reduction with limited risk to salient structure.
- **Deep Layers**: Switch to TPS-style attention criteria to identify salient tokens. By ranking token importance and protecting highly attended tokens from merging, we preserve the concentrated attention patterns required in deep layers.

By explicitly aligning the selection rule with layer-wise attention behavior, SSS preserves deep-layer selectivity while still removing redundancy where it is safe, thereby mitigating token homogenization where selectivity is most critical.

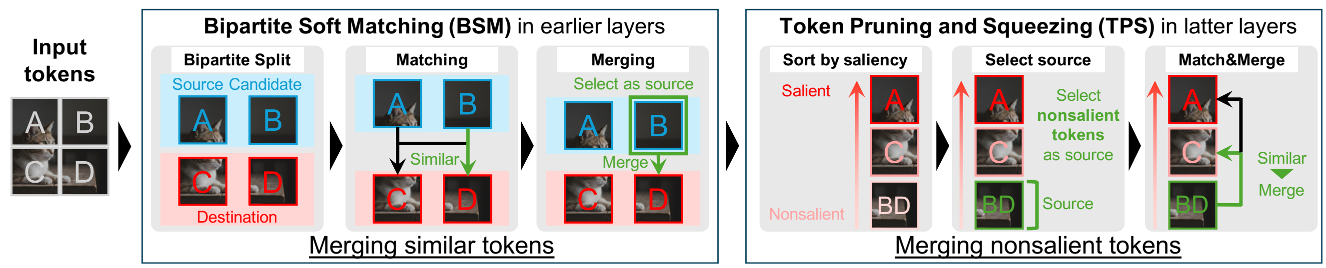


Figure 4. Source Selection Switching (SSS) switches from BSM-style similarity matching in early layers to TPS-style attention-based selection in deeper layers, preserving salient tokens while still merging redundancy.

Algorithm 1 Token Merge with SSS+ME

1: **Input:** tokens $X \in \mathbb{R}^{N \times C}$, kept count K, layer index l, switch depth d, embedding $e^{(l)}_{\text{feat}} \in \mathbb{R}^{C}$
2: **Output:** X_{out}
3: **if** $l < d$ **then** ▷ BSM
4: $(\text{idx}_{\text{src}}, \text{idx}_{\text{dst}}) \leftarrow \text{bipartite\_soft\_matching}(X, K)$
5: **else** ▷ TPS
6: $(\text{idx}_{\text{src}}, \text{idx}_{\text{dst}}) \leftarrow \text{pruning\_and\_squeezing}(X, K)$
7: $\text{src} \leftarrow X[\text{idx}_{\text{src}}]$, $\text{dst} \leftarrow X[\text{idx}_{\text{dst}}]$
8: $\text{src} \leftarrow \text{src} + e^{(l)}_{\text{feat}}$ ▷ apply ME only to sources
9: $X_{\text{out}} \leftarrow \text{scatter\_reduce}(\text{dst}, \text{idx}_{\text{dst}}, \text{src}, \text{reduce} = \text{mean})$
10: **return** X_{out}

4.2. Merge Embedding (ME)

While SSS protects important tokens, merged tokens still undergo averaging, which attenuates token-specific attention-driving components and induces token homogenization. To counteract this effect, we propose **Merge Embedding (ME)**, a lightweight learnable correction that restores these components before reduction. Let C denote the token channel (embedding) dimension. We introduce a per-layer learnable embedding $e^{(l)}_{\text{feat}} \in \mathbb{R}^{C}$ (one C-dimensional vector per layer), trained jointly with the model; this vector is added to source tokens before merging.

As summarized in Algorithm 1, we first choose routing indices with SSS (BSM in early layers, TPS in deep layers), then merge selected sources into destinations via mean aggregation. ME modifies only the *source* side by adding a learnable offset before reduction (line 8), so only tokens merged at that layer receive ME.

Optimizing this embedding through standard training or brief calibration enables the model to regulate the attention scores of merged tokens, with two effects:

1. **Component Restoration**: Restores attention-driving components attenuated by averaging, recovering sharp responses required for concentrated attention.
2. **Influence Regulation**: Prevents merged tokens from either dominating the sequence or being ignored due to weakened signals.

ME is computationally efficient, adding negligible parameters while preserving representational integrity under aggressive token reduction.

5. Evaluation

All evaluations are conducted on ImageNet-1K. We set the SSS switch depth to $d = 2$ for all models based on validation results. For TPS, token importance is computed from head-averaged pre-softmax CLS attention (i.e., the $q_{\text{cls}}k$ score before softmax). ME is initialized to zero and optimized for one epoch using AdamW (learning rate 0.01, no weight decay, batch size 1024) with cross-entropy loss while freezing backbone weights, so that we can isolate the effect of ME.

5.1. SSS: Effect of Layer Threshold

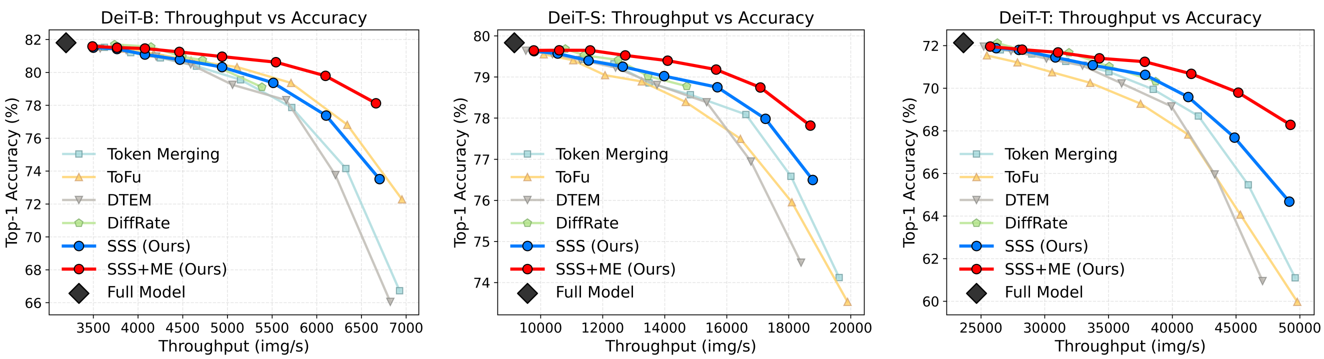
We first evaluate Source Selection Switching (SSS) in isolation by varying the switch depth d that separates similarity-based matching (early layers) from attention-based selection (deep layers). All results in this subsection use DeiT-S on the ImageNet-1K validation set. We control reduction strength through r, the maximum number of tokens merged per layer, following the standard ToMe protocol. Figure 6 reports Top-1 accuracy as a function of d for multiple r values.

Accuracy improves remarkably when the switch occurs after the first few layers, indicating that preserving attention-salient tokens becomes increasingly important in deeper layers. Under mild reduction ($r = 5$), performance remains stable across a broad range of d, and $d = 12$ (equivalent to vanilla ToMe, which always uses similarity matching) remains competitive. Under aggressive reduction ($r = 15$ and $r = 20$), however, the similarity-only strategy degrades rapidly: as depth increases, redundancy is depleted and repeated averaging suppresses tokens that should retain large attention responses.

Using attention-based importance too early is also suboptimal. In high-entropy early layers, attention scores are not yet sufficiently informative, so forcing attention-guided

Model	Method	$r = 8$		$r = 12$		$r = 16$		$r = 20$	
		Acc (%)	Thr (im/s)	Acc (%)	Thr (im/s)	Acc (%)	Thr (im/s)	Acc (%)	Thr (im/s)
DeiT-B	MCTF [12]	81.3	3053	80.6	3655	78.7	4526	70.5	5490
	PPT [24]	81.2	3306	80.3	3879	78.0	4708	65.8	5534
	SSS (Ours)	**81.4**	**3764**	**80.8**	**4468**	**79.4**	**5512**	**73.5**	**6703**
DeiT-S	MCTF [12]	79.4	8063	79.1	9824	78.3	12143	75.0	14624
	PPT [24]	79.4	8385	79.0	9795	78.0	11540	73.1	13170
	SSS (Ours)	**79.6**	**10549**	**79.3**	**12646**	**78.8**	**15700**	**76.5**	**18774**
DeiT-T	MCTF [12]	71.7	20834	71.1	25260	**69.8**	31123	63.1	36666
	PPT [24]	71.6	17100	70.8	19833	68.5	22086	59.1	24168
	SSS (Ours)	**71.8**	**27970**	**71.1**	**33758**	69.6	**41249**	**64.7**	**49173**

Table 1. Comparison of MCTF, PPT, and SSS on DeiT models. Accuracies are rounded to one decimal place.



(a) DeiT-B (base) (b) DeiT-S (small) (c) DeiT-T (tiny)

Figure 5. Accuracy–throughput trade-offs on ImageNet-1K for the DeiT family, comparing token reduction baselines (ToMe [1], Token Fusion (ToFu) [9], DTEM [11], DiffRate [3]) with our SSS and SSS+ME. For all methods except DiffRate, we sweep token-reduction schedules with even r in the range 6–20, while DiffRate is evaluated within the range where training its token-reduction schedule is stable.

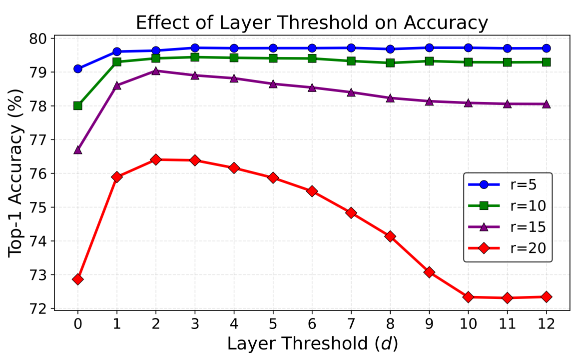


Figure 6. Top-1 accuracy on ImageNet-1K (DeiT-S) as a function of the SSS switch depth d, under different token reduction schedules r. Early layers use BSM and deeper layers use TPS.

merging can reduce accuracy. These results support switching to attention-based selection only after attention becomes reliable, so that low-importance tokens are preferentially merged while dominant late-layer peaks are preserved. Overall, the trend supports our analysis that deep-layer attention is fragile under aggressive merging and that an early switch is critical.

5.2. Comparison with Token Reduction Baselines

We compare our method against representative token-reduction baselines: ToMe [1], ToFu [9], DTEM [11], DiffRate [3], and the recent methods MCTF [12] and PPT [24]. Our variants are SSS and SSS+ME. All experiments are conducted on ImageNet-1K using the DeiT family (Base, Small, Tiny). Throughput (images/s) is measured on a single NVIDIA RTX 4090 (batch size 256, fp16, 200 runs, ‘torch.compile‘ enabled) using PyTorch scaled-dot-product attention (SDPA). In this setup, attention and FFN kernels are highly optimized, making token-merge overhead more visible than in many prior reports.

We report results from two complementary perspectives. Table 1 compares all methods at matched token-reduction schedules, including approaches that are not throughput-competitive in our environment. Figure 5 focuses on throughput-competitive baselines and highlights the practical accuracy–efficiency frontier.

Across all DeiT variants, Table 1 shows that SSS improves throughput while remaining competitive in accuracy

Model	ToMe [1]	SSS	SSS+ME
Accuracy (%)			
CLIP-B [16]	75.7	79.4	**81.3**
MAE-B [8]	54.6	64.4	**78.8**
AugReg-B [18]	63.2	72.5	**76.6**
Throughput (images/s)			
CLIP-B [16]	**6932**	6577	6575
MAE-B [8]	**6904**	6665	6665
AugReg-B [18]	**6902**	6660	6662

Table 2. Broader evaluation on non-DeiT backbones. We use the token-reduction schedule $r = 20$ and compare ToMe, SSS, and SSS+ME on CLIP-B, MAE-B, and AugReg-B.

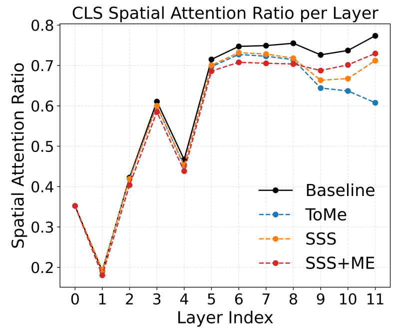


Figure 7. Layer-wise image-to-CLS attention ratio ($r = 20$). SSS/SSS+ME preserve higher late-layer values than ToMe.

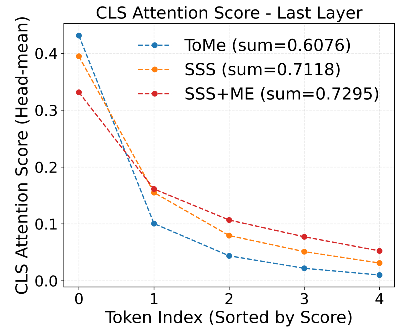


Figure 8. Mean CLS attention over the final five retained image tokens ($r = 20$). ToMe is top-1 dominant, while SSS/SSS+ME are more balanced.

against MCTF and PPT under matched schedules. This trend is amplified in our SDPA-based setup, where merge overhead is exposed more directly. For MCTF, the cost is substantial because it requires the full attention-score matrix and therefore cannot fully exploit the fast SDPA path; in contrast, our CLS-only importance criterion preserves SDPA and only recomputes CLS attention. For PPT, per-instance pruning/pooling switching introduces batch splitting, which further reduces throughput.

Figure 5 shows the accuracy–throughput curves for throughput-competitive baselines. SSS and SSS+ME maintain higher Top-1 accuracy at comparable throughput, and the gap widens as reduction becomes more aggressive. PPT, ToFu, and DiffRate combine merging with pruning and are not the primary target of our study; nevertheless, competitive trade-offs against these methods indicate that mitigating token homogenization is as important as choosing between merging and pruning.

By contrast, MCTF and DTEM primarily refine the matching function: MCTF selects destinations for each source using both similarity and token importance, while DTEM learns a linear token transformation that defines the similarity metric. Relative to these directions, our results highlight a complementary point: layer-wise switching between similarity and token importance is itself a key design factor for effective token matching.

5.3. Broader Model Evaluation

We further evaluate SSS and SSS+ME on models beyond the DeiT family to test whether layer-wise switching and ME generalize across pretraining objectives and backbones. Specifically, we evaluate CLIP-B [16], MAE-B [8], and AugReg-B [18]. Table 2 reports both accuracy and throughput under the token-reduction schedule $r = 20$. SSS consistently improves accuracy over ToMe across all three models, and ME provides additional gains on top of SSS.

5.4. Attention Allocation Analysis

To further validate the mechanism in Sec. 3, we analyze how CLS attention is redistributed when token budgets are highly constrained. We compare ToMe, SSS, and SSS+ME using the statistics shown in Figs. 7 and 8.

First, Fig. 7 reports the layer-wise ratio of CLS attention assigned to image tokens. Consistent with our analysis, ToMe reduces this ratio in later layers, indicating weakened interaction between CLS and visual tokens. In contrast, SSS and SSS+ME maintain higher late-layer ratios, showing that our method better preserves image-to-CLS attention under aggressive reduction.

Second, Fig. 8 examines the final five image tokens in the last attention layer under a schedule of $r = 20$ that remain after applying ToMe-style reduction. We sort these surviving tokens by CLS attention and report their ImageNet-validation averages. Under ToMe, attention is highly concentrated on the top-1 token, while the remaining tokens receive substantially lower attention. With SSS and SSS+ME, this imbalance is reduced, and attention is distributed more evenly across the retained tokens.

These observations suggest that proposed methods utilize the limited tokens more effectively: rather than over-relying on a single dominant token, it preserves multiple informative tokens that remain accessible to the CLS query.

6. Conclusion

We identified **Token Homogenization** as a key limitation of aggressive token merging in ViTs: repeated averaging attenuates attention-driving components and weakens late-layer selectivity. Based on this mechanism, we proposed **Source Selection Switching (SSS)** and **Merge Embedding (ME)**, which respectively adapt merge selection by depth and restore merged-token features before reduction. On ImageNet-1K, SSS and SSS+ME consistently improve the accuracy–efficiency trade-off.

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